Comparing Efficacy of HMMs Trained on Region-based and Function-based Mutations in BRCA1 to Predict Functional Impact

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1 Abstract

This project explores the efficacy of using a single axis of mutation data — location and type — on predicting the pathogenicity of mutations in BRCA1. Due to BRCA1's role as a tumor suppressor gene, we predicted that location-only sorting would create a more accurate predictor than type-only sorting. This was achieved through creating four single-state Hidden Markov Models (HMMs), which were trained on data from ClinVar, and created emissions spreads for: type-based pathogenic mutations, type-based benign mutations, location-based pathogenic mutations, and location-based benign mutations. The log-likelihood scores were compared to create a final type- and location-based prediction of a given mutation's pathogenicity. Final results showed that location-based HMMs were more accurate at predicting mutation pathogenicity than type-based HMMS. These findings indicate that mutation location captures the majority of predictive signal in BRCA1. Single-axis HMM modeling could be extended to other genes to examine similar effects.

2 Introduction and Motivation

Predicting the pathogenicity of a given genetic mutation is highly dependent on a multitude of factors, such as location within the genome, evolutionary conservation, protein structural context, biochemical properties, and molecular consequences. Current computational models, such as REVEL, PolyPhen-2, CADD, and MetaSVM, use many dimensions of data on a gene and mutation in order to make predictions[3]. Because most variant pathogenicity-prediction pipelines use either large neural models or multi-feature ensembles, it remains unclear how much information can be extracted from only a single axis of information, such as a mutation's type or only its genomic location. Hidden Markov Models (HMMs) provide a way to test these features independently through a controlled probabilistic framework.

The importance of a given mutation is not just its location or its type, but its relative impact on the performance of the gene. BRCA1, a tumor suppressor gene[4], was chosen for this comparison due to its critical role in DNA repair. BRCA1 also offers a high volume of documented mutations, as one of the first human genes discovered before the human genome was fully mapped[1]. Due to

BRCA1's role in DNA repair, we know that any mutation (regardless of type) in a critical region is more likely to create a more severe molecular consequence than any mutation in a non-critical region. Therefore, this paper hypothesizes that if two Hidden Markov Models are trained on BRCA1 mutation data, with one using mutation region as the primary feature, and the other using mutation type as the primary feature, then the model using mutation region context will yield a higher classification accuracy for determining pathogenic vs. benign mutations, because the region of a mutation is more strongly associated with functional impact than mutation type.

3 Methods

To test the hypothesis that mutation location is more predictive than mutation type, I developed and trained single-state Hidden Markov Models on BRCA1 mutations.

3.1 Theoretical Basis

The goal of this work is to classify given genetic mutations as benign or pathogenic using Hidden Markov Models. Formally, let the input dataset consist of n mutation samples:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$$

where x_i is a genetic mutation represented by its SPDI description, and $y_i \in \{0, 1\}$ such that a value of 1 denotes pathogenic, and a value of 0 denotes a benign mutation. Based on these data, two different features are considered:

1. Mutation type-based representation, where the mutation is sorted into a discrete biological category.
2. Mutation location-based representation, where the mutation is mapped into a positional bin along the gene.

Two Hidden Markov Models are trained per each feature space. One is trained on benign variants, and the other is trained on pathogenic variants. Expected output is a log-likelihood scoring of a test sample x, classified using maximum likelihood of the outputs of each HMM $\mathcal{M}$:

$$\hat{y} = \arg\max_{c \in \{0,1\}} \log P(x \mid \mathcal{M}_c)$$

This output is given for both the type- and location-based models, and compared in order to determine the overall classification accuracy of each feature.

The algorithm proceeds in four main stages:

1. Data parsing and normalization
2. Feature extraction
3. HMM training
4. Predictive classification

Each HMM has only a single-state, meaning that each model is equivalent to learning a categorical emission distribution per class.

3.1.1 Data Parsing and Feature Extraction

Each mutation is assigned to a discrete biological category and location based on the SPDI field data.

The predefined mapping BIO_MUT_TYPES provides an enumeration of mutation types. More information on this process is discussed in section 3.2.1.

Each mutation is encoded as a one-hot vector:

$$x \in \{0, 1\}^K, \qquad K = |\text{BIO\_MUT\_TYPES}|$$

For each mutation sample's location, each mutation position is projected onto the BRCA1 genomic interval: [BRCA1_START,BRCA1_END), and is then sorted into one of 300 discrete bins representing a uniform location distribution along the gene. The bin size is a tradeoff between spatial resolution and sufficient data per bin. With too low of a bin size, the location data won't accurately find small sections of the gene that are particularly pathogenic, but with too high of a bin size, there may not be any mutation data within a bin, leading to under-fitting.

Each mutation is encoded as a one-hot vector:

$$x \in \{0, 1\}^B, \qquad B = 300.$$

3.1.2 HMM Training

Each model is a Multinomial HMM with one hidden state $N = 1$, and an emission space of K (type-based) or B (location-based). Because there is only one hidden state, the transition matrix is trivially $A = [1]$, and the start probability is: $\pi = 1$. The only learned parameters are the emission probabilities $E = P(x_t \mid z_t)$.

Training is performed using maximum likelihood estimation via the Baum–Welch algorithm as implemented in hmmlearn. With only one hidden state, this simplifies into a frequency-based estimation.

3.1.3 Predictive Classification

Given a test observation x, the log-likelihood under each class model is computed as follows:

$$\ell_0 = \log P(x \mid \mathcal{M}_0), \qquad \ell_1 = \log P(x \mid \mathcal{M}_1).$$

The predicted label is done as follows:

$$\hat{y} = \begin{cases} 1, & \text{if } \ell_1 > \ell_0 \\ 0, & \text{otherwise} \end{cases}$$

A decision tree that used $\geq$ rather than $<$ produced functionally identical results.

3.1.4 Complexity Analysis

Overall time and space complexities are based on: the number of samples, the number of mutation type categories, the number of mutation location categories, and the number of iterations of the Expectation-Maximization (EM) formula.

Let:

- n be the number of samples,
- K be the number of mutation-type categories,
- B be the number of location bins (300),
- $T = 1$ be the sequence length per sample,
- I be the number of EM iterations (50),
- $N = 1$ be the number of states in the HMM.

All constants (K, B, T, I, N) are independent of n.

Time Complexity: For data loading and parsing, each SPDI string is parsed and normalized once in constant time, leading to a total time complexity of $O(n)$. To construct each feature's one-hot encoding, each sample takes $O(K)$ time for type-based and $O(B)$ time for location-based encodings, leading to an overall time complexity of:

$$O(nK) + O(nB).$$

During Multinomial HMM training, Baum–Welch (used by hmmlearn) has the time complexity:

$$O(I \cdot N^2 \cdot T \cdot D),$$

Where D is the emission alphabet size. Because $N = 1$ and $T = 1$ in our models, this simplifies to:

$$O(I \cdot D).$$

Translating this to each HMM model, the type-based model training has time complexity $O(I \cdot K)$, and the location-based model training has time complexity $O(I \cdot B)$.
During evaluation, Each test sample requires one likelihood evaluation under the benign HMM, and one likelihood evaluation under the pathogenic HMM. Each likelihood computation is linear, leading to a time complexity of:

$$T_{\text{eval}} = \begin{cases} O(nK) & \text{(location-based)} \\ O(nB) & \text{(type-based)} \end{cases}$$

Summing all the steps, we get total time complexities for both models.

$$\begin{aligned} T_{\text{type}} &= O(n) + O(nK) + O(IK) + O(nK) \\ &= O(nK + IK) \end{aligned}$$

$$\begin{aligned} T_{\text{location}} &= O(n) + O(nB) + O(IB) + O(nB) \\ &= O(nB + IB) \end{aligned}$$

When the program is run with both type and location-based models (as is the default setting), the models are executed sequentially, leading to a time complexity of:

$$\begin{aligned} T_{\text{total}} &= O(nK + IK) + O(nB + IB) \\ &= O(n(K + B) + I(K + B)) \end{aligned}$$

Since I, K, and B are constants with respect to n — that is, they don't change depending on the input dataset's size — we can simplify the total time complexity of this project's code into simply $O(n)$.

Space Complexity: The size of the loaded data is created with respects to n, as each sample creates a stored parsed SPDI object and label, leading to a space complexity of $O(n)$. The feature matrices for both models are stored both in relation to n and in relation to K or B, depending on the model's feature of interest, leading to a space complexity of $O(n \cdot D)$. For each HMM, the start probabilities are $(O(1))$, transition matrix are $(O(1))$, and emission probabilities are $(O(D))$. When both models are executed, we get an overall space complexity of:

$$\begin{aligned} S_{\text{total}} &= O(n + nK + nB + 1 + 1 + D) \\ &= O(n) \end{aligned}$$

These complexity calculations show that the single-state HMM framework of this project scales linearly with any given dataset size, which is ideal for larger genomic datasets and makes this project easy to replicate and expand upon, as processing power and data storage are not severe limitations compared to more complex predictive models.

3.2 Implementation Details

The implementation of this study was done entirely in Python, using the hmmlearn library for Hidden Markov Model construction and training, scikit-learn for dataset splitting and performance evaluation, and NumPy and Pandas for data handling. The code is structured in a way that maximizes reproducibility, clarity, and expansion into use with other genomic datasets.

Data parsing and handling was achieved in the DataPreparation class, including reading data from given files, filtering data, and parsing into mutation type and location bins for training and evaluation by HMMs.

TypeHMM and LocationHMM implement their respective classifiers. Each model creates, trains, and tests two HMMs each for benign and pathogenic data. Both classes share a consistent interface that ensured a fixed random seed and shared training/testing splits. The main program allows for customization and flexibility in execution, such that custom benign and pathogenic data may be inputted, and users can select which models to run. Both models will be run by default in order to compare model accuracy.

Data structures used: Python dictionaries are used for SPDI parsing and mutation data; NumPy arrays are used for one-hot encoding emissions, as required by hmmlearn's multinomial HMM class[2]; and lists store per-sample sequences and sequence lengths, also as required by hmmlearn.

3.2.1 Challenges and Design Tradeoffs

A key challenge was classifying mutation types for the model. Although the ClinVar dataset included a BIO_MUT_TYPES column, many mutations had no data in that column, requiring custom logic to be implemented in order to classify mutations by type. Because of this, several categories of mutation listed in the enum are unused because they are not generated by the custom classification rules. Only the following categories are actually produced by the classifier:

- Frameshift
- In-frame indel
- Microsatellite

- Missense
- Structural CNV

Although other defined categories exist in the enumeration mapping, they are never emitted, due to constraints on the information and logic required to produce accurate classifications of certain categories.

Another constraint was the use of single-state HMMs. While using a single state limits the modeling power, it also allows for a simpler interpretation. Utilizing a multiple-state HMM would increase time and model complexity, while also causing the models to utilize sequential dependencies, which is not necessary for this project, as these data exist only as single-observation sequences.

4 Experimental Design and Datasets

The datasets used in this study consist of two downloads of clinical data from ClinVar, hosted by the National Center for Biotechnology Information[5]. Both files are of genetic mutations associated with the BRCA1 gene; the benign file has 813 mutations, and the pathogenic file has 4049 mutations. Only variants that fall within the given gene locus are used in the location-based model, in order to ensure that each mutation can be accurately binned. All entries were explicitly classified as either ”benign” or ”pathogenic,” and all entries were marked as reviewed in order to ensure accuracy. The dataset is split randomly into a training set and testing set, where 70% of the data are used to train the HMMs, and 30% of the data are used to test the models. Because there are significantly more pathogenic entries than benign entries (due partially to clinical interest in BRCA1 leading to a higher degree of sequencing of pathogenic mutations), stratified sampling is used to preserve the benign/pathogenic class ratio and avoid under- or over-fitting models during training. Stratified sampling preserves the underlying ratio of benign to pathogenic mutations, which works to prevent training bias during model evaluation.

4.1 Validation

A stratified train/test split ensures that there is no leakage between the training and testing phases of each HMMs lifecycle, and maintains balanced class proportions for training and testing models.

For individual variants, the log-likelihoods of benign or pathogenic status are calculated by both HMMs in each feature.

To verify that the models learned biologically meaningful patterns, the patterns of each HMM’s generated emission matrices were compared to the actual dataset’s proportions of mutation data, as seen in Figures 2–5. In addition, the overall predictions of pathogenicity by each HMM pair were graphed in relation to the actual dataset.

5 Results

Performance of models was assessed using accuracy, precision, recall, and F1 scores for both models. The location-based model has a 91% accuracy, precision, and F1-score, with a 92% recall score, which indicates that location data provided a strong and accurate basis for predicting pathogenicity. The type-based model performs moderately well (accuracy 77%, precision 88%, recall 77%, F1-score 79%), but shows reduced recall, suggesting missed pathogenic variants. The performance gap between the two sets of models supports our hypothesis that BRCA1 mutation location is a stronger predictor of pathogenicity than mutation type. Confusion matrices were used to analyze error and possible class imbalances.

Table 1: Confusion matrix for the mutation-type-based model.

	Predicted Benign	**Predicted Pathogenic**
Actual Benign	232	11
Actual Pathogenic	287	766

The high false negatives for pathogenic variants, also shown through the low recall score, indicates that the type-based model has a limited ability to differentiate between pathogenic and benign mutations, in such a way that pathogenic mutations are over-predicted. The particular miss-classification of actual pathogenic variants as benign likely reflects that multiple mutation types (e.g., missense vs frameshift) can occur across the same critical regions, making it harder to discriminate pathogenicity from type alone.

Table 2: Confusion matrix for the location-based model.

	Predicted Benign	**Predicted Pathogenic**
Actual Benign	160	83
Actual Pathogenic	25	1028

With the location-based model, we can see substantially fewer false negatives than in the type-based model. The location-based model provides a more sensitive measurement of pathogenic mutations, which is especially important if used for clinical screening of new, unclassified mutations.

5.1 Visuals

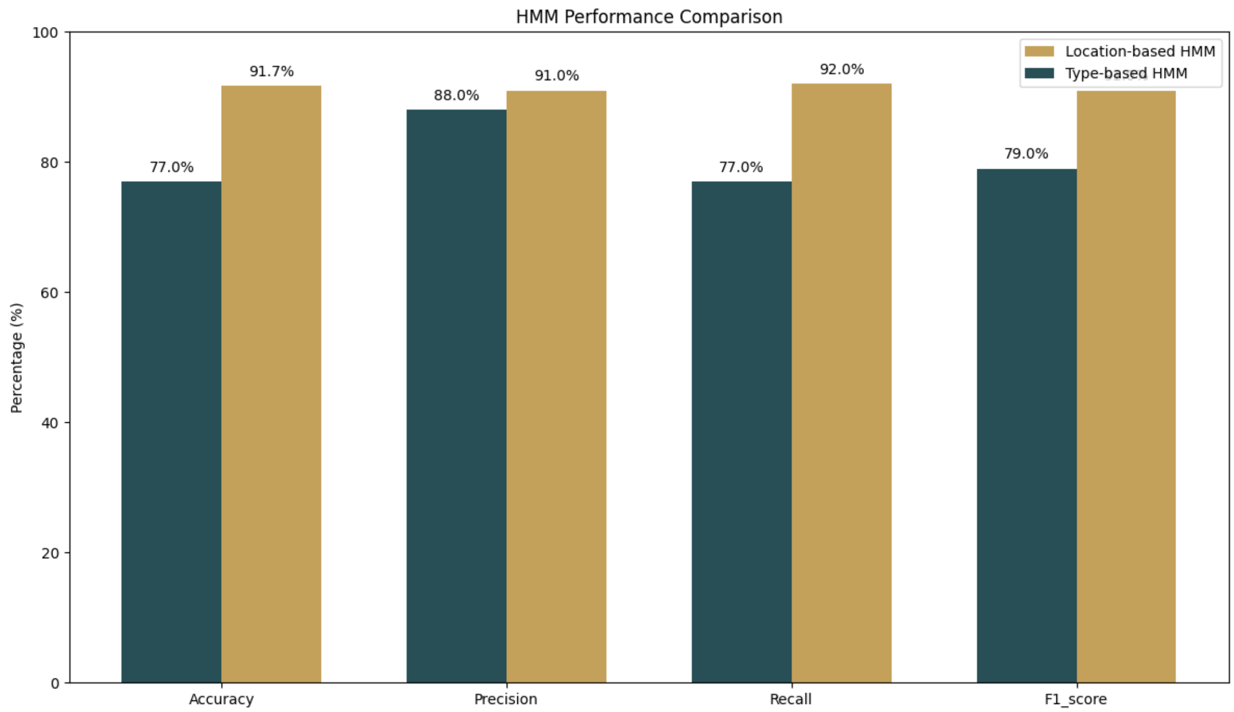

Figure 1: Performance comparison between location-based and type-based models.

5.1.1 Correctness

Manual checks of emission matrices confirmed that the emission probabilities closely align with the actual dataset, as shown in Figures 2–5.

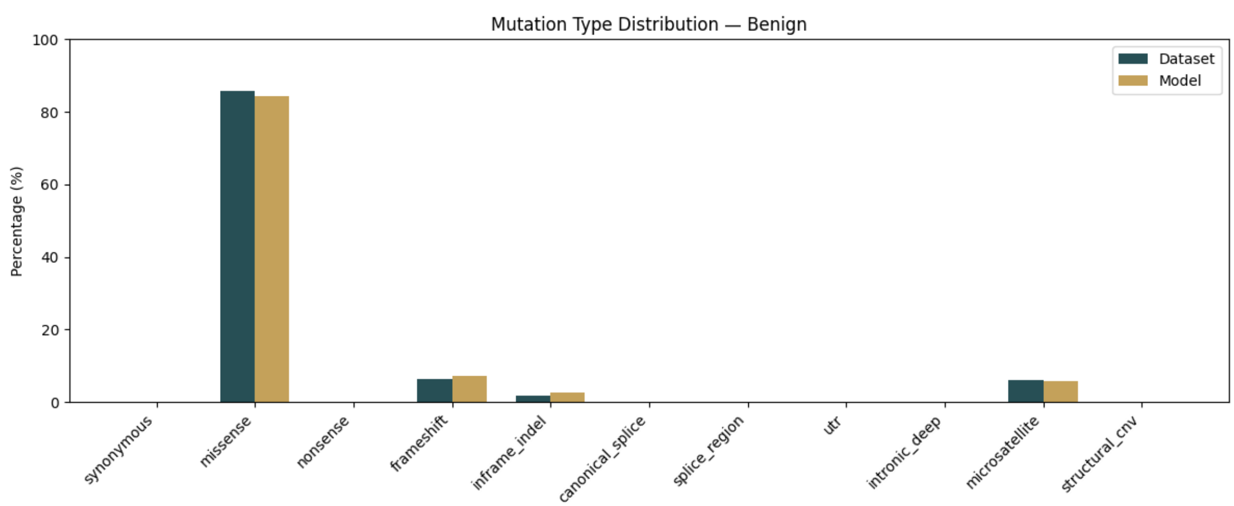

Figure 2: Emission probabilities of the the type-based benign HMM compared to actual dataset proportions.

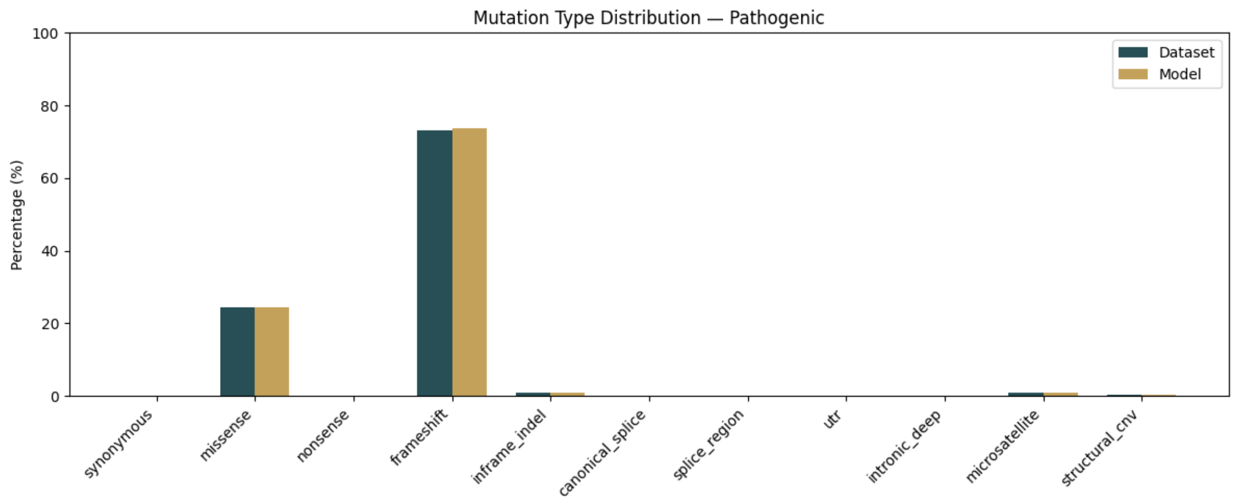

Figure 3: Emission probabilities of the type-based pathogenic HMM compared to actual dataset proportions.

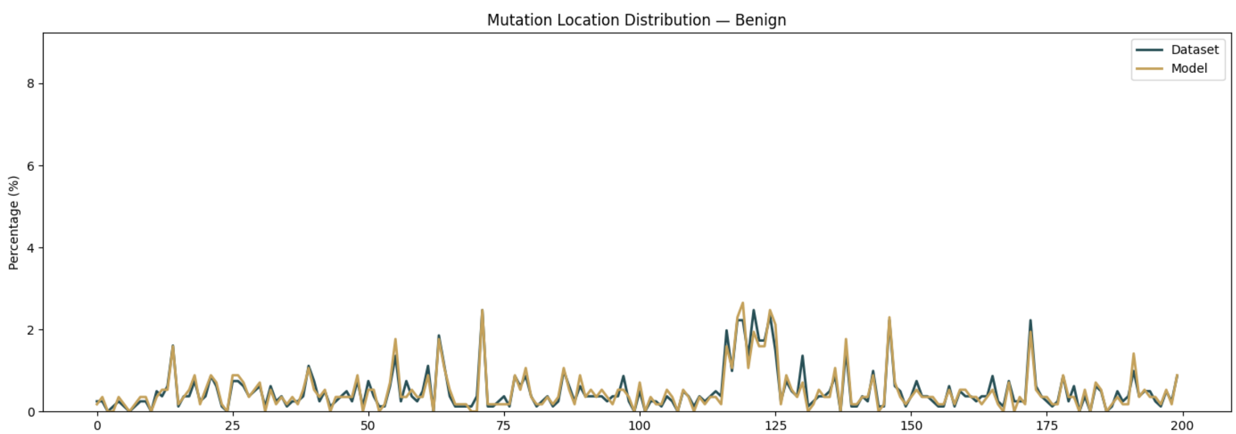

Figure 4: Emission probabilities of the location-based benign HMM compared to actual dataset proportions.

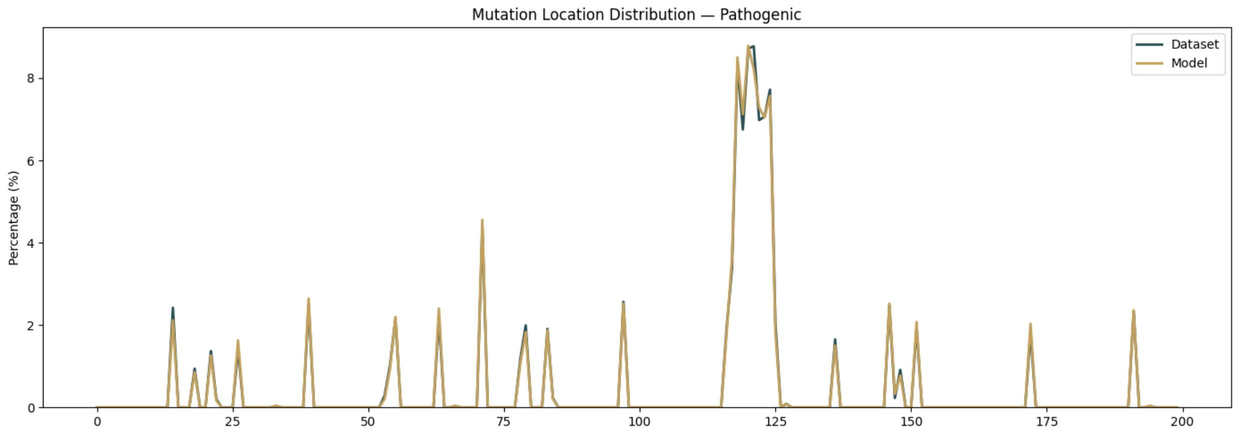

Figure 5: Emission probabilities of the location-based pathogenic HMM compared to actual dataset proportions.

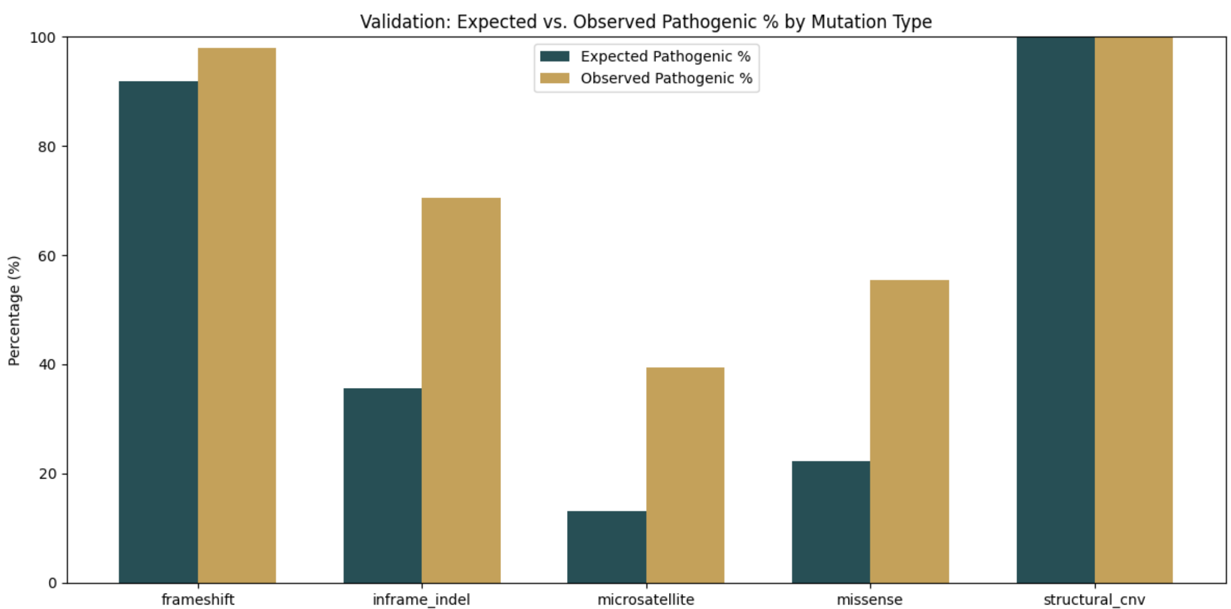

Figure 6: Comparison between predicted pathogenicity and actual pathogenicity with the type-based model.

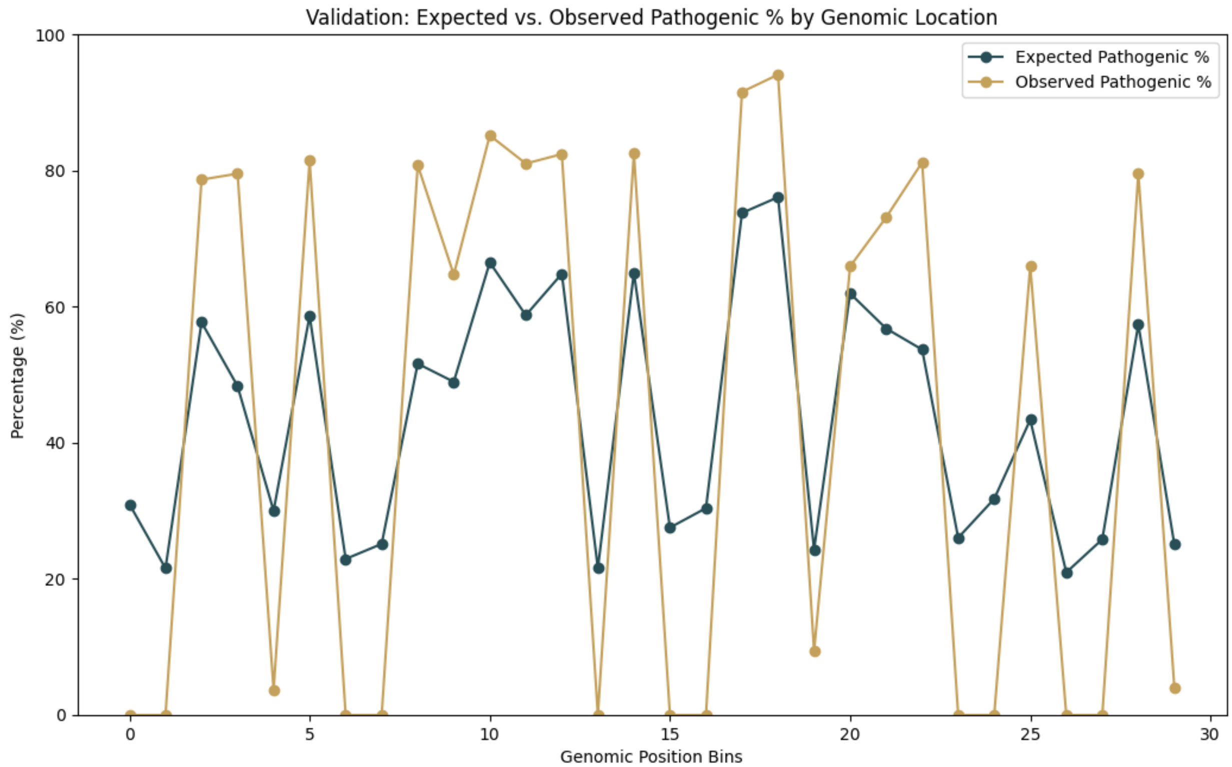

Figure 7: Comparison between predicted pathogenicity and actual pathogenicity with the location-based model.

6 Discussion and Future Work

The results of this study demonstrate a clear advantage of using location-based mutation data over type-based data for predicting the pathogenicity of BRCA1 mutations. The location-only HMM achieved an overall accuracy of 91.7%, compared to 77% for the type-only model. Moreover, the location-based model exhibited substantially higher recall for pathogenic variants (0.98 vs 0.73), highlighting its ability to more reliably identify clinically significant mutations. This aligns with the biological understanding that BRCA1's functional domains are highly sensitive to disruption, and that mutations in critical regions are more likely to have deleterious effects, regardless of the precise mutation type. The type-only model demonstrated lower recall and a higher number of false negatives for pathogenic variants. This indicates that mutation type alone is insufficient for capturing the context-dependent functional impact in BRCA1, where the positional context is the dominant determinant of pathogenicity. While type-based information may still provide valuable insight—particularly for missense versus nonsense discrimination—its predictive power is weaker when used in isolation.

A current limitation of this model is in the custom mutation type classifier, which is very coarse at the moment, especially compared to the 300-bin location classifier. Developing a more robust and complex classifier of mutation type may increase predictive power of type-based models. In addition, data imbalance skewed toward pathogenic data means that the models trained on benign-only data may be less predictive due to data limitations rather than model limitations.

Future work includes applying this methodology to other genes with differing functional architectures, such as regulatory genes or oncogenes. BRCA1 is a tumor suppressor gene where loss-of-function mutations will have the most pathogenic impact, which is primarily location-based. This single-axis modeling system may draw different conclusions in different regulatory genes where loss-of-function is less important, and the pathogenic impact may depend more on mutation type or other molecular properties than strict location.

Additionally, extending the model to incorporate additional biological features—such as evolutionary conservation, protein structural domains, or splicing impact—could improve classification for genes where pathogenicity is more multifactorial. Third, while single-state HMMs were sufficient for this study due to the single-observation nature of the data, multi-state HMMs could be explored for genes with sequentially dependent functional regions, such as exons spanning multiple structural domains.

References

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